

LEGEND

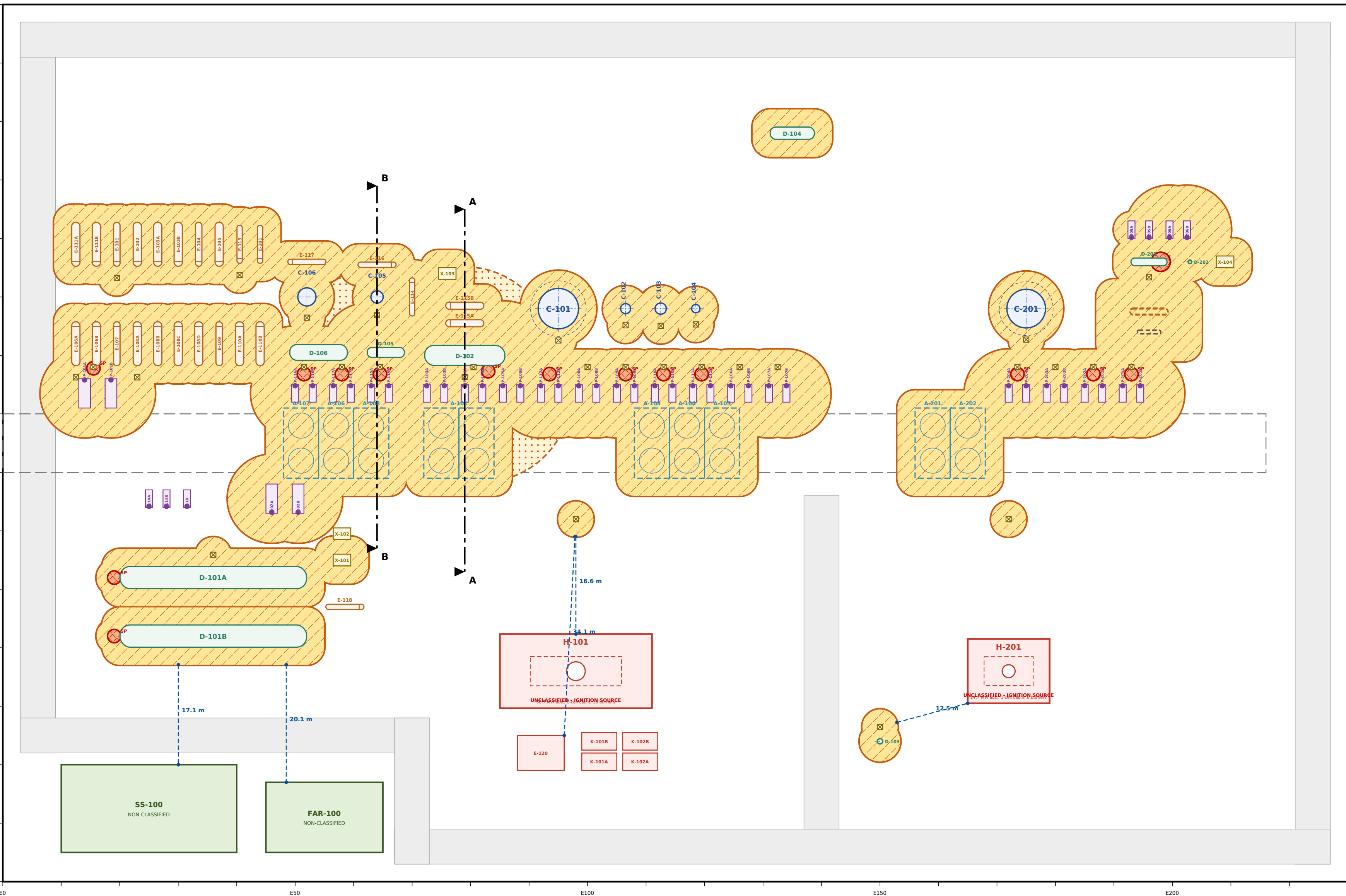
- ZONE 1 - flammable atmosphere likely in normal operation (primary-grade sources)
- ZONE 2 - not likely in normal operation; short duration if it occurs (secondary grade)
- ADDITIONAL ZONE 2, 0.6 m above grade (cat. A / highly volatile liquids, 15 m)
- NON-CLASSIFIED BUILDING (pressurised) / unhatched area = unclassified
- CV station (indicative) SP sample point V atmospheric vent - - - dimension to nearest Zone 2

FLUID CATEGORIES

CAT.	FLUID CATEGORY (EI 15)
A	flammable liquid which on release vaporises rapidly and substantially (LPG, unstabilised naphtha)
B	flammable liquid at a temperature sufficient for boiling on release (crude, naphtha)
C	flammable liquid that may be above its flash point or form a mist on release (kerosene, diesel, gas oils, residue)
G	refinery fuel gas / off-gas (may contain H ₂ , H ₂ S)
N	not a release source

RELEASE SOURCE SUMMARY (full schedule CFU-000-EL-HAC-003)

SOURCE TYPE	CAT	GRP	ZONES	EXTENT	EQUIPMENT
Atmospheric vent	G	IB	Zone 1/Zone 2	1.5 m; 4.5 m	D-201
Control-valve station	A	IA	Zone 2	3 m	C-105, P-104, P-115
Control-valve station	A	IB	Zone 2	3 m	D-102
Control-valve station	B	IA	Zone 2	3 m	C-101, C-106, D-101, P-101, P-116, P-117
Control-valve station	C	IA	Zone 2	3 m	C-102, C-103, C-104, C-201, E-101, E-106, E-108, E-113, H-101...
Control-valve station	G	IB	Zone 2	3 m	D-103, D-201
Ejector / condenser flanges	G	IB	Zone 2	3 m	J-201, 203 / E-202, 204 (ST-201)
Exchanger flanges	A	IA	Zone 2	3 m	E-114
Exchanger flanges	B	IA	Zone 2	3 m	E-116, E-117
Exchanger flanges	B	IB	Zone 2	3 m	E-115A, E-115B
Exchanger flanges	C	IA	Zone 2	3 m	E-101, E-102, E-103A, E-103B, E-104, E-105, E-106A, E-106B, E-...
Header box plugs / flanges	A	IA	Zone 2	3 m	A-106
Header box plugs / flanges	B	IA	Zone 2	3 m	A-107, A-108
Header box plugs / flanges	B	IB	Zone 2	3 m	A-101
Header box plugs / flanges	C	IA	Zone 2	3 m	A-103, A-104, A-105, A-201, A-202
Pump seal	A	IA	Zone 2/Zone 2 ext	15 m; 7.5 m	P-104, P-115
Pump seal	B	IA	Zone 2	7.5 m	P-101, P-102, P-103, P-116, P-117
Pump seal	C	IA	Zone 2	7.5 m	P-106, P-107, P-108, P-109, P-110, P-111, P-112, P-201, P-202...
Pump seal	C	IB	Zone 2	3 m	P-105, P-205
Pump seal	C	IB	Zone 2	7.5 m	P-206
Sample point	A	IA	Zone 1/Zone 2	1 m; 3 m	P-115
Sample point	A	IB	Zone 1/Zone 2	1 m; 3 m	D-102
Sample point	B	IA	Zone 1/Zone 2	1 m; 3 m	D-101A, D-101B, P-101, P-116, P-117
Sample point	C	IA	Zone 1/Zone 2	1 m; 3 m	P-109, P-110, P-111, P-112, P-201, P-202, P-204
Vessel flanges / instruments	A	IA	Zone 2	4.05 m	C-105
Vessel flanges / instruments	A	IA	Zone 2/Zone 2 ext	15 m; 7.5 m	D-105
Vessel flanges / instruments	A	IB	Zone 2/Zone 2 ext	15 m; 7.5 m	D-102
Vessel flanges / instruments	B	IA	Zone 2	3 m	D-101A, D-101B, D-106
Vessel flanges / instruments	B	IA	Zone 2	4.55 m	C-106
Vessel flanges / instruments	B	IA	Zone 2	6.45 m	C-101
Vessel flanges / instruments	C	IA	Zone 2	3 m	X-101, X-103, X-104
Vessel flanges / instruments	C	IA	Zone 2	3.7 m	C-104
Vessel flanges / instruments	C	IA	Zone 2	3.85 m	C-102
Vessel flanges / instruments	C	IA	Zone 2	3.95 m	C-103
Vessel flanges / instruments	C	IA	Zone 2	6.3 m	C-201
Vessel flanges / instruments	G	IB	Zone 2	3 m	D-104, D-201
Vessel flanges / instruments	G	IB	Zone 2	3.3 m	D-202
Vessel flanges / instruments	G	IB	Zone 2	3.45 m	D-103



SCALE 1:500 (A1) - 0 10 20 30 40 50 m

CLEARANCE OF IGNITION SOURCES / BUILDINGS TO NEAREST CLASSIFIED AREA

UNCLASSIFIED ITEM	CLEARANCE m	NEAREST SOURCE	DESCRIPTION	CHECK
H-101	16.6	RS-130	Control valves FV-1011, FV-1012, FV-1013, FV-1014, FV-1015,	OK
H-201	12.5	RS-124	Control valves LV-9002 (D-103)	OK
E-120	34.1	RS-130	Control valves FV-1011, FV-1012, FV-1013, FV-1014, FV-1015,	OK
SS-100	17.1	RS-076	D-101B Electrostatic desalter, 2nd stage	OK
FAR-100	20.1	RS-076	D-101B Electrostatic desalter, 2nd stage	OK

MINIMUM EQUIPMENT PROTECTION (IEC 60079-14 / NEC 505)

EQUIPMENT	ZONE 1	ZONE 2
LV motors (800 V)	Ex db or Ex eb (IE, stall relay), IIB T3 (Gb)	Ex ec (non-sparking), IIA/IB T3 (Gc); AEx ec per NEC 505
MV motors (4.16 kV)	Not located in Zone 1 (relocate) / Ex pxb	Ex ec with stator discharge risk assessment / pre-start purge, IIA T3
VFD-fed motors	Ex db certified with the drive (converter duty)	Ex ec certified for converter duty; T-class verified with VFD
Instruments	Ex ia / ib (intrinsic safety), Ex db	Ex ia / ic, Ex db, Ex ec
Lighting fittings	Ex db / eb, IIB T3	Ex ec / nL, IIA T3 (IIB in H2S areas)
Junction boxes / glands	Ex eb / ec boxes; Ex db barrier glands for Ex d	Ex eb / ec boxes and glands
Heat tracing	Ex 60079-30-1, stabilised design <= T3	Ex 60079-30-1, self-regulating, T3

NOTES

- Classification per API RP 505 (Zone system) and NFPA 70 Art. 505; release-source extents from RP 505 figures for heavier-than-air vapours in open, adequately ventilated (natural) areas; fluid categories per EI 15 (Model Code of Safe Practice P 15).
- Point-source method (no blanket classification). Areas not shown hatched are unclassified.
- Gas group IIA, temperature class T3 generally; IIB where H₂S / fuel gas / off-gas is present (overhead system D-102 / A-101 / E-115, hotwell D-201, D-202, D-103, D-104, ejector set, sour-water pumps).
- Fired heaters H-101 / H-201 are ignition sources and are not classified; fuel-gas valve trains are of welded construction with minimum flanges; electrical equipment within 3 m of FG flanges shall be Zone 2 rated. Distances from the nearest Zone 2 boundary are dimensioned.
- PSVs discharge to the closed flare system - no atmospheric release envelope. Drains are closed (to OWS / sour water); below-grade pits, trenches and catch basins within a Zone 2 are Zone 1.
- Pipe rack: only flanges, valves and instrument connections are sources (3 m Zone 2); straight welded runs are not sources. Control-valve stations are shown at indicative locations pending P&IDs / piping layout.
- Pump seals are secondary-grade releases (API 682 seals, open area) -> Zone 2 only; Zone 1 applies to primary-grade sources (sample points, try-cocks, atmospheric vents).
- SS-100 and FAR-100 are non-classified pressurised buildings; HVAC intakes shall be located in unclassified area (>= 3 m from any Zone 2 boundary). Gas detection at intakes.
- Equipment surface temperatures of process plant may exceed fluid AIT (hot oil > 260 C); T-class applies to electrical equipment. Release source schedule: CFU-000-EL-HAC-003; sections: CFU-000-EL-HAC-002.

REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
HAZARDOUS AREA CLASSIFICATION PLAN - ZONE 1 / ZONE 2 (API RP 505)			
DISCIPLINE	STAGE	SCALE	SHEET
ELECTRICAL	FEED	1:500	1 OF 1
DRAWN CLAUDE CODE	CHECKED -	APPROVED -	DATE 2026-10-02
DRAWING NUMBER CFU-000-EL-HAC-001			REV A